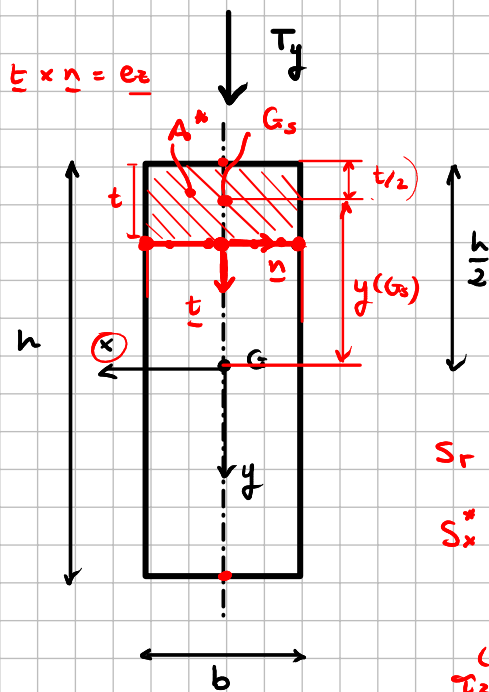




$$\tau_m = \tau_{zt,m} = - \frac{T_y}{s I_x} S_x^*$$

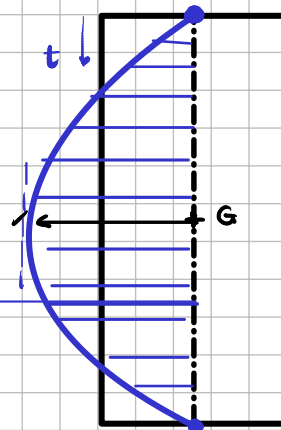
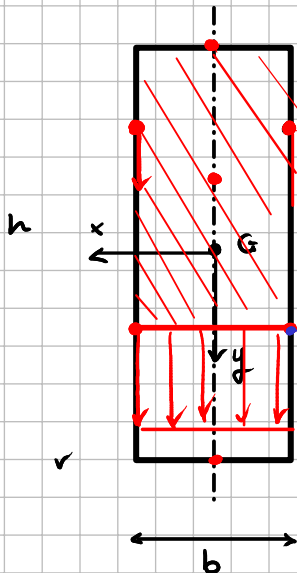
S_x^* = momento statico della porzione di sezione retta "alle spalle" del versore tangente $\underline{t}$ alla linea media

$$S_r = A \, d_r G$$

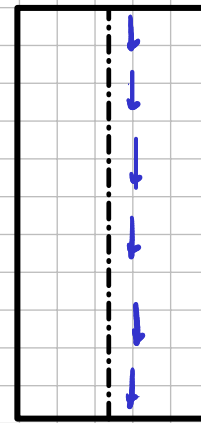
$$S_x^* = A^* \, d_x G_s = A^* y(G_s) = -bt \left(\frac{h}{2} - \frac{t}{2} \right) \propto t^2$$

$$\tau_{zt}^{(\tau)} = - \frac{T_y}{s I_x} \left(-bt \left(\frac{h}{2} - \frac{t}{2} \right) \right)$$

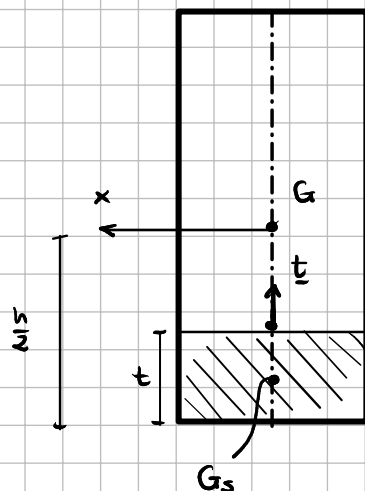
$$\begin{aligned} \frac{d\tau_{zt}^{(\tau)}}{dt} &= + \frac{T_y}{s I_x} \left(b \left(\frac{h}{2} - \frac{t}{2} \right) - \frac{bt}{2} \right) = \\ &= \frac{T_y}{s I_x} \left(\frac{bh}{2} - bt \right) \stackrel{!}{=} 0 \quad t = \frac{h}{2} \end{aligned}$$



Moduli

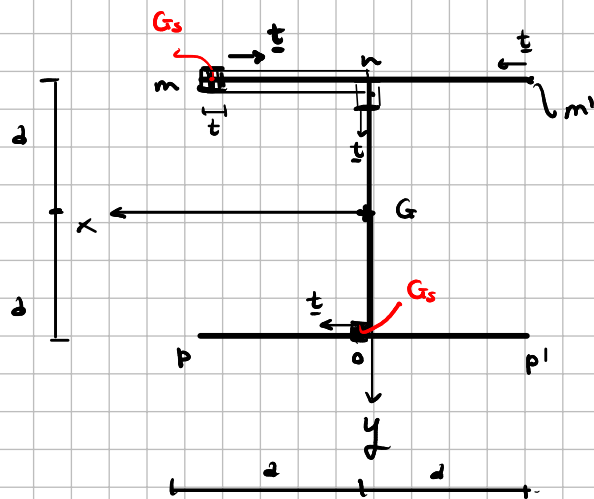


Segni



$$S_x^* = bt \left(\frac{h}{2} - \frac{t}{2} \right)$$

$$\tau_{zt} = - \frac{T_y}{s I_x} S_x^*$$



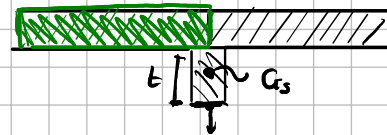
• sezione simmetrica $\rightarrow T_{zt}^{(1)}$ speculari rispetto all'asse di symm

• $S_{x,mn}^* = -st(a) \propto t$

$$S_{x,no}^* = S_{x,mn}^*(t=a) + S_{x,m'n}^*(t=a) + S_{x,n}^* =$$

$$= 2S_{x,mn}^*(t=a) + S_{x,n}^* =$$

$$= -2sa^2 + S_{x,n}^* = -2sa^2 - st(a - \frac{t}{2}) \propto t^2$$



$$S_{x,op}^* = \frac{S_{x,no}^*(t=2a)}{2} + st(a) \propto t$$

